Assignment 1 Report

1. LogMelFilterBanks implementation

Implemented in `melbanks.py`: - Hann window (`torch.hann_window`) - STFT with power spectrum (`power=2.0`) - Mel projection via `torchaudio.functional.melscale_fbanks` - Output as `torch.log(mel_spec + 1e-6)`

2. Validation vs torchaudio

Validation script:

```
python3 scripts/check_logmel.py --wav-path data/SpeechCommands/speech_commands_v0.02/yes/0a7
```

Observed: - shape match: `(1, 80, 101)` - `torch.allclose(...) = True` - max absolute diff = `0.0`

Plots: - `artifacts/plots/logmel_comparison.png` - `artifacts/full/plots/logmel_comparison_real_wav`

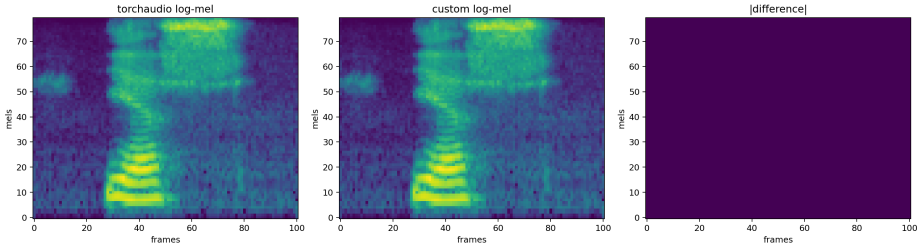


Figure 1: LogMel comparison (real wav)

3. Binary yes/no pipeline

- Dataset: `torchaudio.datasets.SPEECHCOMMANDS`
- Classes used: `yes`, `no`
- Splits: default `training / validation / testing`
- Audio preprocessing: mono, resample to 16 kHz, pad/truncate to 16000 samples

4. Model and complexity

Model: `SpeechYesNoCNN` (Conv1d-based, with LogMel features).

Parameter budget check: - all trained variants are `< 100K` parameters - range observed: `43,905` to `63,105` params

FLOPs are measured via forward hooks (Conv1d + Linear) and written to `summary.csv`.

5. Training setup

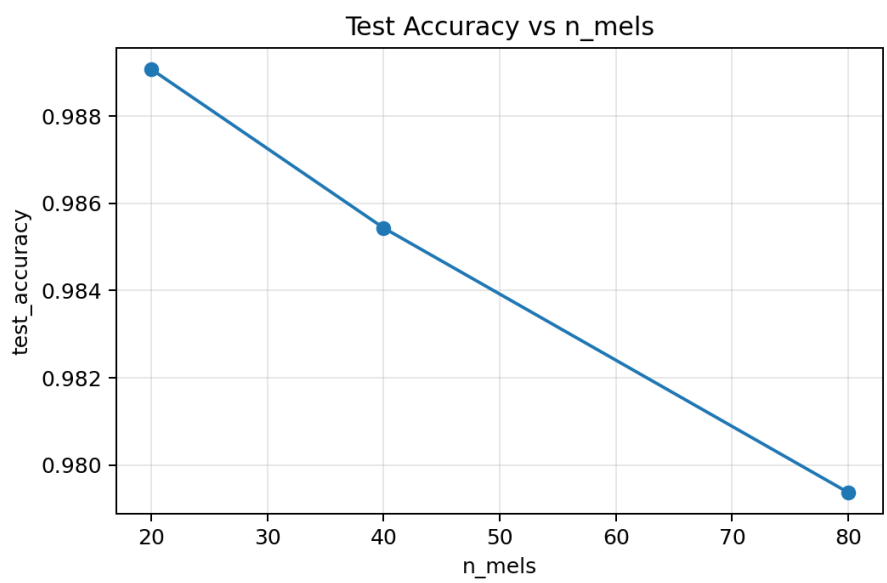
- Optimizer: Adam
- Loss: BCEWithLogitsLoss
- Epochs per run: 3
- Logged per epoch:
  - `train_loss`
  - `val_accuracy`
  - `epoch_time_sec`
- Test metric:
  - `test_accuracy`

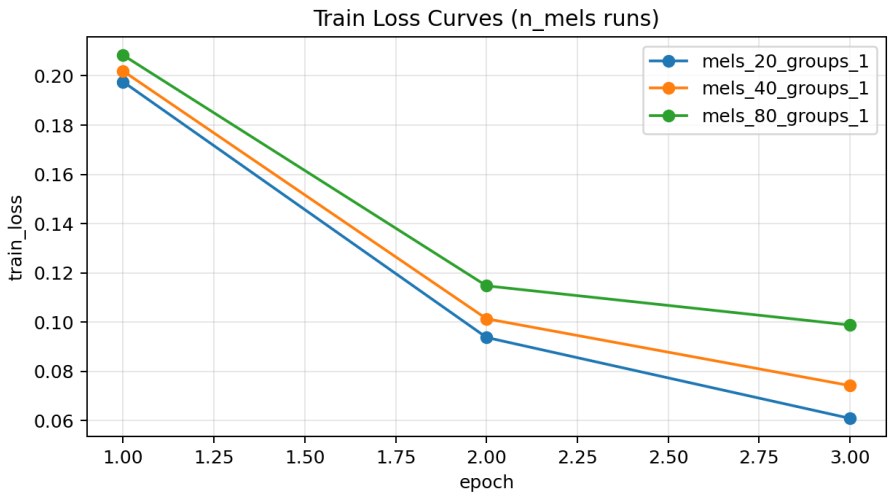
Artifacts: - histories: `artifacts/full/logs/*_history.csv` - checkpoints: `artifacts/full/checkpoints/*.pt` - summary: `artifacts/full/summary.csv`

6. n_mels experiments

run	n_mels	test accuracy	best val accuracy	mean epoch time (s)
mels_20_groups_1	20	0.9891	0.9788	17.400
mels_40_groups_1	40	0.9854	0.9851	17.226
mels_80_groups_1	80	0.9794	0.9751	13.945

Plots: - `artifacts/full/plots/n_mels_vs_test_accuracy.png` - `artifacts/full/plots/train_loss_n_me`



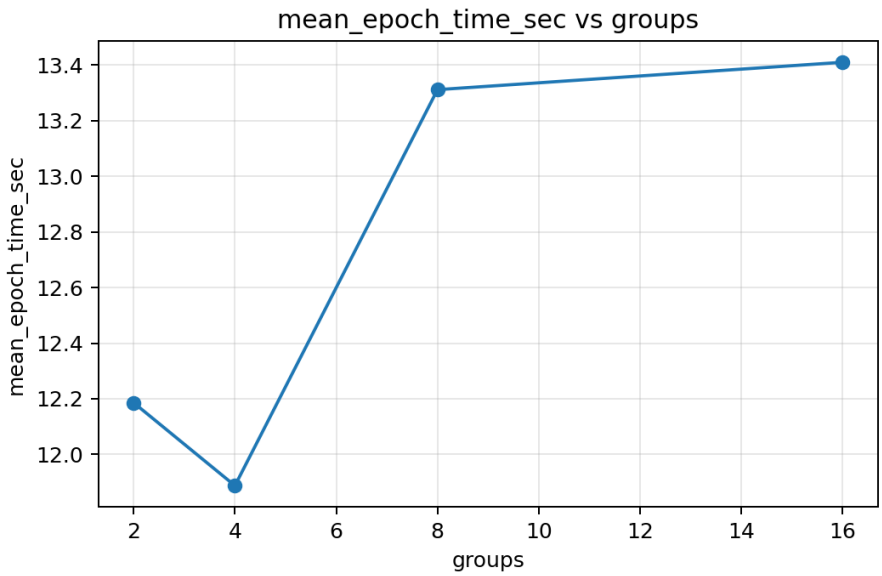


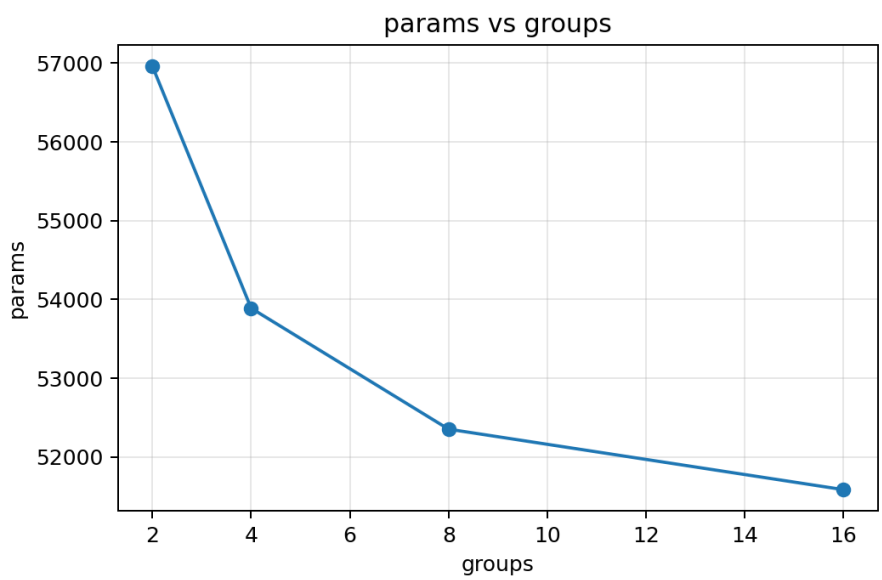
Observation: - Best test accuracy in this setup: `n_mels=20`. - `n_mels=80` was fastest by epoch, but had lower test accuracy than 20/40.

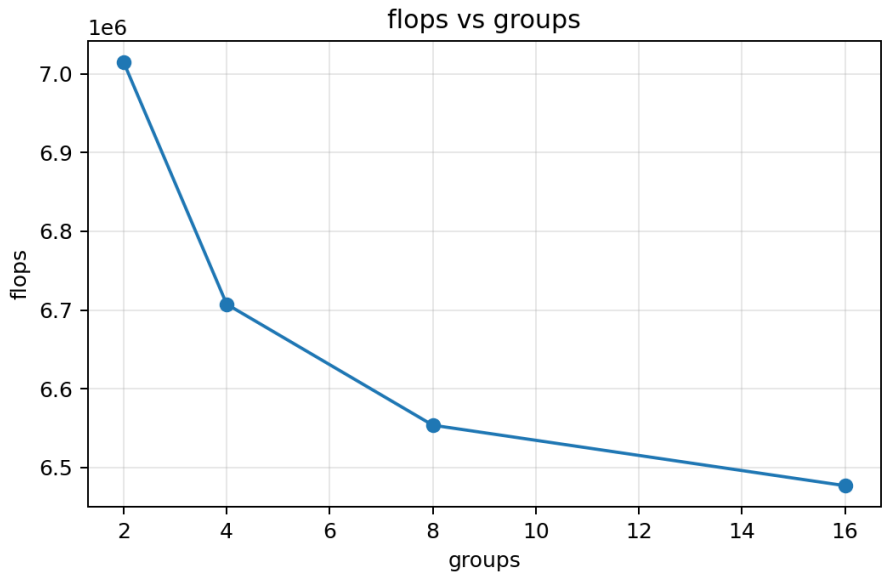
7. groups experiments (baseline n_mels=80)

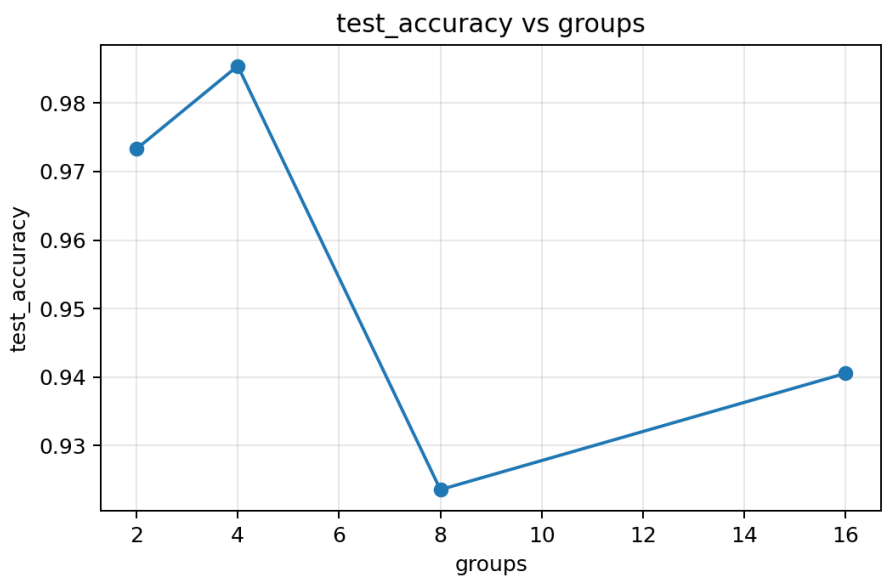
run	groups	params	FLOPs	test accuracy	best val accuracy	mean epoch time (s)
mels_80_groups_2	2	56,961	7,014,656	0.9733	0.9813	12.186
mels_80_groups_4	4	53,889	6,707,456	0.9854	0.9851	11.887
mels_80_groups_8	8	52,353	6,553,856	0.9235	0.9751	13.312
mels_80_groups_16	16	51,585	6,477,056	0.9405	0.9676	13.410

Plots: - `artifacts/full/plots/groups_vs_mean_epoch_time_sec.png` - `artifacts/full/plots/groups_vs_params.png` - `artifacts/full/plots/groups_vs_flops.png` - `artifacts/full/plots/groups_vs_test_accuracy.png`


mean_epoch_time_sec vs groups
mean_epoch_time_sec
13.4
13.2
13.0
12.8
12.6
12.4
12.2
12.0
2
4
6
8
10
12
14
16
groups


params vs groups
params
57000
56000
55000
54000
53000
52000
2
4
6
8
10
12
14
16
groups





Observation: - Increasing `groups` reduces params/FLOPs. - Best quality among tested groups is `groups=4`. - Very high grouping (`8`, `16`) reduced quality noticeably in this setup.

8. Final conclusions

- Custom `LogMelFilterBanks` is numerically equivalent to torchaudio MelSpectrogram + log.
- Conv1d binary classifier meets `<100K` params constraint for all tested configs.
- For this 3-epoch setup:
  - best overall test accuracy: `mels_20_groups_1` (`0.9891`)
  - best grouped-conv trade-off: `mels_80_groups_4` (`0.9854`)
- Suggested baseline for next stage:
  - `n_mels=20, groups=1` if maximizing accuracy
  - `n_mels=80, groups=4` if prioritizing grouped-conv efficiency with strong quality